

**Determining the Top Predictors of Taglish Election Misinformation
Through a Comparative Detection and Corpus Analysis Using
RoBERTa-Tagalog, mBERT, and XLM-RoBERTa**

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Chapter I.

THE PROBLEM AND ITS BACKGROUND

A. Project Context

The Philippines consistently ranks among the countries with the highest social media engagement globally, with an estimated 95.8 million active users comprising 81.9% of its total population as of October 2025 [1]. Platforms such as Facebook and X (formerly Twitter) have become the primary channels through which millions of Filipinos access news and public information. Within this digital landscape, misinformation spreads rapidly and pervasively, often taking the form of Taglish, a code-switched variety of Filipino and English that reflects the everyday communication patterns of Filipino internet users.

Unlike formal Filipino or standard English, Taglish is characterized by its informal and inconsistently structured nature, presenting significant challenges for natural language processing (NLP) systems [2]. Election periods amplify this concern considerably, as politically charged content and unverified claims circulate widely across platforms in the weeks surrounding major electoral events. The 2022 Philippine presidential election, in particular, was marked by widespread reports of election-related misinformation spreading through social media, where much of it appeared in code-switched Taglish discourse [3], [4].

While transformer-based fake news detection systems have demonstrated strong performance across various global contexts, many are designed primarily for English or formal text and may exhibit reduced performance when applied to code-switched, informal content such as Taglish [5]. Prior studies in Filipino NLP have largely focused on structured datasets derived from formal news registers, rather than the informal social media discourse where misinformation is most prevalent [6], [7]. Since the introduction of the BERT pretraining framework, transformer-based architectures have been successfully applied to fake news detection across a number of languages [8], [9], [10], [11]. However, a focused comparative evaluation of RoBERTa-Tagalog, mBERT, and XLM-RoBERTa specifically on Taglish election misinformation remains a gap in the

literature. To the researchers' knowledge, no prior study has directly compared these three transformer models on Taglish election misinformation classification.

Beyond the question of which model performs best, a further gap exists in understanding what these models actually respond to — specifically, which textual features most strongly drive their classification decisions on Taglish election content. Political entities, candidate names, election-related terms, and informal Taglish expressions may all serve as influential features in model classifications, yet this dimension of analysis has received little attention in the Filipino NLP literature. This study addresses both gaps by comparatively evaluating three pre-trained transformer models on Taglish election misinformation detection and conducting a model interpretability analysis to identify and examine the top predictors shaping each model's classification behavior.

B. Purpose and Description

This study evaluates which pre-trained transformer model — when fine-tuned on Filipino election misinformation data — performs best at classifying Taglish election-related posts as credible or not credible, and identifies whether political entities, candidate names, election-related terms, and informal Taglish expressions emerge as the linguistic features strongly influencing the models' classification decisions.

To accomplish this, the study employs three integrated components: a comparative experimental design, a descriptive corpus analysis, and a model interpretability analysis. The comparative experimental component evaluates and benchmarks three transformer-based language models — RoBERTa-Tagalog, mBERT, and XLM-RoBERTa — fine-tuned on a combined primary corpus drawn from existing Filipino misinformation datasets. Model performance is assessed on both an internal formal test set and a supplementary Taglish presidential election social media test set consisting of 300 to 500 manually collected, cleaned, and labeled public posts from Facebook and X pertaining to the 2022 Philippine presidential election period, specifically from February 8 to May 9, 2022.

The descriptive corpus analysis examines the structure, thematic content, political entities, issue categories, and code-switching patterns of the supplementary dataset prior to model evaluation, establishing a transparent empirical baseline against which the



interpretability findings are interpreted. The model interpretability component then applies token-level attribution methods — Integrated Gradients as the primary method and SHAP as a cross-validation approach — to identify and rank the textual features most strongly influencing each model’s classification decisions, with particular attention to whether political entities, candidate names, election-related terms, and informal Taglish expressions consistently emerge as influential predictors across models.

C. Objectives of the Study

This study aims to evaluate RoBERTa-Tagalog, mBERT, and XLM-RoBERTa in classifying Taglish election-related posts as credible or not credible, and to identify whether political entities, candidate names, election-related terms, and informal Taglish expressions serve as influential features in the models' classifications. Specifically, this study seeks to:

1. Fine-tune and obtain benchmark scores from three transformer models — RoBERTa-Tagalog, mBERT, and XLM-RoBERTa — on a combined Filipino misinformation dataset for binary credibility classification.
2. Develop a supplementary Taglish election-related test dataset from publicly available Facebook and X posts, repost captions, and eligible threads, which will be manually collected, cleaned, and labeled to evaluate real-world model performance on informal Taglish election discourse.
3. Assess and contrast how the models perform on both the supplementary Taglish election dataset and the internal formal test set. Using standard NLP metrics—specifically precision, recall, accuracy, and F1-score—this evaluation will identify which architecture is most effective at processing the unique linguistic properties of Taglish.
4. Investigate the consistency with which election-related terminology, candidate names, political entities, and informal Taglish phrases appear among the primary attribution features across all tested transformer models through the application of SHAP and Integrated Gradients.
5. Analyze patterns of candidate visibility, political targeting, and election-related misinformation reflected in the dataset's influential features based on the top predictor rankings and corpus-level findings.

Through this five-objective framework, the study contributes both a comparative technical benchmark for Filipino NLP and an interpretable account of the linguistic and political features associated with Taglish election misinformation.

D. Significance of the Study

This study addresses a gap in Philippine misinformation research by providing a comparative performance benchmark and an interpretable account of the linguistic and political features associated with Taglish election misinformation, carrying relevance across multiple stakeholder groups.

For Filipino internet users, the study may inform future tools that provide credibility assistance for Taglish election-related content, encouraging critical media consumption.

For the Filipino NLP research community, the study will contribute to one of the few direct comparative benchmarks of transformer models specifically focused on Taglish misinformation classification. The supplementary Taglish-labeled election test dataset may also support future research on Filipino NLP, code-switching, misinformation detection, and explainable AI.

For journalists and fact-checkers, the findings may support the future development of assistive screening tools for assessing the credibility of social media content, especially during election-related monitoring, reporting, and investigation.

For educators and media literacy advocates, the study may provide examples of how misinformation is expressed through informal Taglish, political claims, and candidate-centered narratives. These findings may be useful in teaching responsible online information sharing.

For policymakers and election-related institutions, the study may provide insights into the patterns of Taglish election misinformation and the political entities commonly associated with it. The study does not aim to predict electoral results but rather to analyze patterns of visibility, political targeting, and misinformation in the collected dataset.

From a technological standpoint, the use of transformer-based models alongside token-level attribution methods offers a more transparent approach to misinformation detection, one that identifies specific textual features influencing each model's classification decisions rather than relying solely on accuracy scores, thus making the classification process transparent and beneficial for further research.

Furthermore, this research aligns with national development targets, specifically regarding digital innovation, quality education, democratic involvement, and the cultivation of informed communities. By supporting United Nations Sustainable Development Goals such as Goal 4 (Quality Education), Goal 9 (Industry, Innovation, and Infrastructure), Goal 11 (Sustainable Cities and Communities), and Goal 16 (Peace, Justice, and Strong Institutions), the project advances technological progress, ethical information exchange, and more robust public conversation within the digital landscape of the Philippines.

E. Scope and Limitation

1. Scope

The scope of this study is defined by the following conditions:

Three Transformer Models. The study focuses on the comparative evaluation of three pre-trained transformer models: RoBERTa-Tagalog, mBERT, and XLM-RoBERTa. These are fine-tuned for binary credibility classification of Taglish social media content.

Binary Credibility Classification. Each post in the supplementary dataset is classified into one of two categories: credible or not credible.

2022 Philippine Presidential Election Period. The study is confined to publicly accessible content from Facebook and X pertaining to the 2022 Philippine presidential election period, which runs from February 8 to May 9, 2022, corresponding to the official campaign period through election day.

300–500 Supplementary Taglish Election Test Set. The supplementary dataset to be constructed will comprise 300 to 500 manually collected, cleaned, and annotated posts from publicly accessible Facebook and X content.



Primary Training Corpus. The primary training corpus is drawn from two existing labeled Filipino misinformation datasets: the Fake News Filipino Dataset [6] and the Philippine Fake News Corpus [7]. This serves as the supervised learning resource for model fine-tuning, not as a basis for evaluating real-world performance.

2. Limitations

The study acknowledges the following limitations inherent to its design and scope.

300–500 Supplementary Test Set. The number of posts may not fully represent the broader population of Taglish election misinformation that circulated during the 2022 Philippine presidential election period, potentially limiting the top predictor rankings' generalizability beyond the collected dataset.

Research Team as Annotators. The use of research team members as primary annotators introduces the possibility of familiarity bias; however, this risk is mitigated through independent labeling and a documented inter-annotator agreement procedure requiring a minimum Cohen's Kappa of $\kappa \geq 0.61$ before considering that the dataset is finalized.

Integrated Gradients and SHAP as Approximations. The token-level attribution methods provide approximations of model behavior. Findings from the interpretability analysis should therefore be understood as patterns observed within the collected dataset under the specific fine-tuning conditions of this study.

Formal Filipino Training Corpus. The primary training corpus consists of formal Filipino news text rather than informal, code-switched Taglish social media content. While this domain gap is an intentional design feature of the study, it enables the assessment of out-of-domain generalization, it nonetheless constitutes a limitation in that the models are not trained on data that is linguistically representative of the supplementary test set.

Resource-Dependent SHAP Subsampling. SHAP analysis will be applied to a stratified random subsample of the interpretability subset as its computation over transformer-based models is more resource-intensive than Integrated Gradients. The subsample size will be determined by available computational resources and documented transparently in the final manuscript.

F. Conceptual Framework

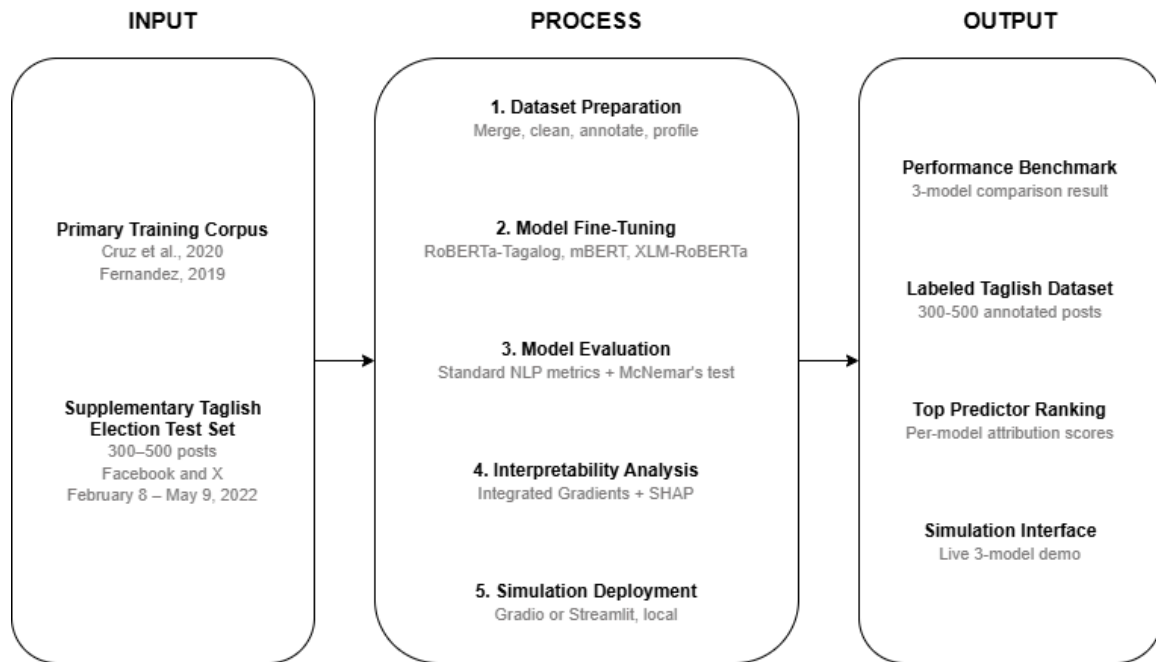


Fig 1. Conceptual Framework of the study

The conceptual framework of this study is organized around an input-process-output (IPO) structure that reflects the sequential and interconnected nature of its three research components: comparative model evaluation, corpus analysis, and model interpretability analysis. A simulation-based demonstration is additionally produced as a supplementary proof-of-concept output.

The framework utilizes two distinct input streams. Initial fine-tuning is supported by the primary training corpus, which integrates the Fake News Filipino Dataset and the Philippine Fake News Corpus to provide necessary labeled data. In contrast, the second input consists of a novel supplementary Taglish test set. This resource contains 300 to 500 social media posts related to the 2022 Philippine presidential election, manually gathered and annotated from public Facebook and X content. Serving as a unique empirical element, this supplementary dataset is used solely for evaluation to determine how effectively each model generalizes to the informal, code-switched discourse found in real-world election-related misinformation.

These two inputs feed into five sequential process phases. In the first phase of dataset preparation, the primary corpora are merged, cleaned, deduplicated, and stratified into



training, validation, and internal test sets, respectively, utilizing a 70/15/15 ratio, while the supplementary Taglish test set is organized in terms of collection and screening, annotation through Cohen's Kappa inter-annotator agreement measurement, entity and issue tagging, and exploratory corpus profiling. For the second phase, model fine-tuning, three pre-trained transformer models (RoBERTa-Tagalog, mBERT and XLM-RoBERTa) are independently fine-tuned on the combined primary corpus based on identical hyperparameter configurations for a binary credibility classification task so as to conduct a controlled and unbiased comparison over all three architectures. In the third phase, model evaluation — each fine-tuned model is evaluated across two test levels, namely the internal formal test set and the supplementary Taglish presidential election test set, using accuracy, F1-score, precision, recall and confusion matrices as the assessment tools, with the supplementary test set serving as the primary indicator of real-world generalization. In the fourth phase, interpretability analysis, token-level attribution methods are applied to explain model decision-making on the supplementary Taglish test set. Integrated Gradients serves as the primary attribution method, while SHAP is used as a cross-validation interpretability approach. The resulting influential features are ranked and categorized according to political entities, election-related terms, Taglish expressions, and issue keywords, then compared across all three models. In the fifth phase, simulation-based deployment, the three fine-tuned model checkpoints are loaded into a locally executed interface that accepts a manually entered Taglish social media post as input and returns, in real time, the predicted credibility label, confidence score, and top attribution tokens for all three models displayed side by side, making comparative model behavior tangible and demonstrable beyond aggregate evaluation metrics.

The framework produces four outputs. The first is a comparative performance benchmark of all three transformer models on Taglish election misinformation classification, the first of its kind in Filipino NLP literature. The second is the labeled Taglish social media dataset of 300 to 500 annotated posts, which constitutes a reusable foundation for future Filipino misinformation research. The third is a set of ranked top predictor attributions per model, providing an interpretable account of the linguistic and political cues each model associates with election misinformation in Taglish. Furthermore, a locally executable simulation interface is developed to showcase the comparative classification and attribution behavior of the three fine-tuned models when processing live Taglish input.



This serves as a practical proof-of-concept for accessible, real-time misinformation assessment.

Collectively, these elements establish a unified research pipeline that transitions from raw data acquisition to thorough model comparison and interpretability analysis, culminating in a functional simulation demo. This approach provides a technical benchmark for Filipino NLP while offering a concrete proof-of-concept for the detection of Taglish election misinformation.

G. Operational Definition of Terms

Binary Credibility Classification - The process of categorizing each Taglish election-related post into one of two labels, credible or not credible.

Candidate Visibility - The frequency with which a candidate or political figure appears in the supplementary Taglish election dataset, corpus-level summaries, or top predictor rankings. In this study, candidate visibility refers only to observed presence in the collected dataset and does not imply support, intent, causality, or electoral influence.

Claim Type — Refers to the specific category of a factual statement published by a user, which may manifest as fabricated quotes, distorted statistics, fictitious occurrences, manipulated contexts, or unsupported accusations.

Code-Switching (Taglish) - This refers to the linguistic practice of alternating between two languages—specifically Filipino (or Tagalog) and English—within a single social media post, whether during a specific sentence or throughout the entire discourse.

Cohen's Kappa - The statistical measure used to assess agreement between annotators beyond chance. In this study, a minimum value of $\kappa \geq 0.61$ is required before the dataset is considered finalized.

Corpus Analysis - The systematic examination of the supplementary Taglish election dataset to identify language patterns, commonly used terms, candidate references, political entity names, and informal language present in credible and not credible posts.

Credible Post – Taglish posts on an election topic that make a factual statement, which is verified or substantiated by credible sources.

Election-Related Terms – Specific words or phrases associated with topics related to elections, such as vote buying, endorsing, canvassing, disqualification, legitimacy, and voting.

Influential Features – Features that the models have found influential in determining classification decisions. Examples include key political organizations, candidate names, election-related terms, and informal Taglish phrases.

Informal Taglish Expressions - Slang, meme phrases, hashtags, conversational expressions, and informal mixed language words commonly found in Taglish social media posts.

Integrated Gradients – This is the primary method for interpretability employed in this research to assess the contribution of individual words or tokens to the classification decisions made by the model.

Issue Category – This pertains to the central election-related subject of a post, such as election legitimacy, historical claims, false endorsements, misleading statistics, or voting fraud.

mBERT – A multilingual transformer-based model built on the BERT framework and pre-trained using Wikipedia data from 104 languages to handle multilingual NLP tasks. In this research, it is fine-tuned to perform binary credibility classification on election-related posts in Taglish.

McNemar's Test – A statistical evaluation used to confirm if the differences in performance between two models are statistically significant rather than resulting from random chance.



Not Credible Post – An election-related Taglish post featuring factual claims that are unsupported, manipulated, misleading, false, or otherwise refuted by verifiable and dependable sources.

Political Entities – The various political institutions and actors mentioned within the gathered Taglish posts, encompassing candidates, parties, figures, the COMELEC, judicial bodies, and other election-related organizations.

Political Targeting – A recurring trend in the dataset where specific political figures, candidates, or institutions are linked to misleading content, misinformation claims, or particular issue-based narratives.

RoBERTa-Tagalog – A transformer-based model pre-trained on extensive Tagalog and Filipino text to identify language-specific patterns for Filipino NLP. It is fine-tuned in this study for the binary credibility classification of election-related Taglish content.

SHAP - The secondary interpretability method that computes Shapley values to approximate the contribution of each token to a model's classification decision. In the study, it is used to cross-validate the token-level attribution scores produced by Integrated Gradients.

Supplementary Taglish Presidential Election Test Set - The manually collected dataset of 300 to 500 public Taglish social media posts from Facebook and X related to the 2022 Philippine presidential election. This dataset is used only for external model evaluation and top predictor extraction.

Taglish – The type of language used in the Philippines where Filipino/Tagalog and English words are mixed in one discussion or message on social media, in one sentence.

Taglish Election Misinformation - Social media content related to the election in Taglish that contains a factual claim about the 2022 Philippine presidential election and is found to be false, misleading, manipulated, unsupported or contradicted by reliable verification sources such as COMELEC, Vera Files, Tsek.ph, or credible news reports.



Top Predictors - The particular words or textual markers with the highest cumulative attribution score from model classification and those with the strongest association with either credible or non-credible posts in the additional Taglish data.

Verifiable Claim - Statement in the post that can be validated through reliable sources; any post without verifiable claims, such as slogans or opinions, is not part of the study.

XLM-RoBERTa - A language model based on a transformer architecture that was extensively trained on large datasets from 100 languages. This is used for cross-lingual natural language processing tasks. Specifically, XLM-RoBERTa is adapted in the current study for Taglish posts' credibility classification.

Chapter II.

REVIEW OF RELATED LITERATURE AND STUDIES

A. Taglish and Code-Switching in NLP

Code-switching is the act of switching between two or more languages within a single utterance, sentence, or conversation. It is a common feature of multilingual communication, especially in digital and social media settings. In the Philippines, this phenomenon manifests primarily as Taglish, a hybrid of Filipino or Tagalog and English that dominates daily online discourse. While code-switching reflects natural bilingual communication patterns among Filipino internet users, it poses considerable challenges for NLP systems that are typically trained on monolingual or formally structured text.

Herrera, Aich, and Parde [12] introduced TweetTaglish, a large-scale dataset of more than 20,000 Tagalog-English code-switched posts collected from Twitter, with a specific goal to facilitate NLP research on Taglish. Their work addressed the significant underrepresentation of Tagalog-English in NLP research and documented that even basic tasks such as word-level language identification are substantially more difficult in code-switched settings, particularly when alternating languages share borrowed vocabulary. TweetTaglish is most directly comparable to the supplementary Taglish dataset that will be constructed in the present study, highlighting the importance of purpose-built, domain-specific resources for Filipino NLP.

Nguyen and Bryant [13] developed CanVEC, a Vietnamese-English code-switching natural speech corpus, highlighting that code-switching resource construction in Southeast Asian bilingual communities requires domain-specific collection strategies tailored to the particular language pair and communicative context. Their work demonstrated that structural alternation patterns between a regional language and English vary substantially across different Southeast Asian language pairs, reinforcing that multilingual or monolingual resources cannot simply be repurposed for code-switched settings without significant loss of linguistic validity. This cross-linguistic parallel directly supports the present study's rationale for constructing a purpose-built Taglish election dataset rather than adapting existing resources, as the informal, election-specific



nature of Taglish social media discourse requires a similarly targeted collection and annotation approach.

Cruz and Cheng [2] developed RoBERTa-Tagalog and other large-scale Filipino language models that demonstrate language-specific pre-training on Filipino and Tagalog corpora, resulting in significantly improved performance on Filipino NLP tasks compared to using general multilingual models. They also highlighted the persistent gap between formal Filipino text and informal, code-switched social media content — the same domain gap that the present study intentionally exploits to assess out-of-domain generalization across three transformer architectures.

Nazir, Faisal, Habib, and Ahmad [14] applied mBERT to multiclass sentiment analysis of code-mixed Roman Urdu and Roman Punjabi data, where substantial performance improvement was found over benchmark algorithms. Their study demonstrated that multilingual transformer models fine-tuned on code-mixed, low-resource data can capture linguistic patterns that monolingual models trained on standard text cannot, which is a finding that directly informs the evaluation of the present study of mBERT's classification behavior on Taglish content.

Sheth, Sinha, Patil, Beniwal, and Singh [15] surveyed over 300 studies on code-switching-aware NLP in the era of large language models and discovered that most models continue to face challenges with mixed-language inputs because of limited code-switched training data and evaluation biases favoring monolingual benchmarks. Their survey identified informal Taglish as a particularly underrepresented domain, directly contextualizing the research gap addressed by the present study.

The reviewed literature establishes that Taglish and code-switched text more broadly pose persistent challenges for NLP systems built on monolingual assumptions. Available resources remain limited in scope and domain specificity, and existing transformer models underperform on code-switched content relative to monolingual counterparts. The present study addresses this gap by constructing a purpose-built Taglish election dataset and evaluating three transformer models specifically on this informal, code-switched domain.



B. Fake News and Misinformation Detection Using Transformer Models

Since the introduction of BERT, transformer-based architectures have become the dominant paradigm for automated fake news and misinformation detection across a wide range of languages and domains. This section reviews studies applying transformer models to credibility classification, with attention to comparative evaluations, fine-tuning strategies, and performance on informal or low-resource text.

Devlin, Chang, Lee, and Toutanova [8] introduced BERT, establishing the foundational transfer learning paradigm for NLP classification tasks. Their demonstration that pre-trained representations can be fine-tuned with minimal task-specific modifications directly motivates the fine-tuning approach adopted in the present study, wherein pre-trained transformer models are adapted for binary credibility classification through the addition of a classification output layer.

Conneau et al. [16] introduced XLM-RoBERTa, a multilingual transformer pre-trained across 100 languages, demonstrating strong cross-lingual transfer performance that outperformed mBERT on several multilingual benchmarks. The model's capacity for cross-lingual generalization makes it particularly relevant for Taglish classification, where the model must process a code-switched variety not explicitly represented in its pre-training data.

Kaliyar, Goswami, and Narang [9] developed FakeBERT, a BERT-based approach for binary fake news detection on social media, establishing the viability of the transformer fine-tuning paradigm for credibility classification. Their architecture — attaching a classification head to a pre-trained BERT encoder — directly mirrors the model adaptation strategy used in the present study.

Raza and Ding [5] proposed a transformer-based fake news detection system integrating news content and social context features, noting that code-switched and informal content remains a significant challenge for detection systems trained on formal text. This observation directly motivates the present study's use of a supplementary Taglish test set as a real-world out-of-domain evaluation environment.

Raja, Soni, and Borgohain [10] applied transfer learning with adaptive fine-tuning to fake news detection in Dravidian low-resource languages, demonstrating the feasibility of adapting multilingual transformer models for non-English misinformation classification



— a result that supports the approach adopted in the present study for the Filipino NLP context.

Raza, Paulen-Patterson, and Ding [11] compared BERT-family models and large language models for fake news detection, finding that fine-tuned BERT-class models remain competitive for binary classification tasks. Their comparative methodology, evaluating multiple models under the same experimental conditions, closely mirrors the controlled evaluation design of the present study.

The reviewed literature demonstrates strong transformer-based performance on English fake news detection but reveals a consistent absence of comparative evaluations on code-switched, informal content. No prior study has directly benchmarked RoBERTa-Tagalog, mBERT, and XLM-RoBERTa on Taglish election misinformation — a gap the present study directly fills.

C. Filipino NLP and Dataset Development

The development of NLP resources for Filipinos has progressed significantly in recent years, yet substantial gaps remain, particularly in the availability of informal, code-switched, and domain-specific datasets. This section reviews the foundational datasets and language model contributions that directly inform the present study's training corpus, model selection, and evaluation design.

Cruz, Tan, and Cheng [6] presented the Fake News Filipino Dataset, comprising approximately 3,206 labeled Filipino news articles and representing one of the earliest standardized benchmarks for Filipino misinformation classification. Their multitask transfer learning framework demonstrated that English NLP models can be effectively localized to Filipino text through fine-tuning. This dataset serves as one of the two primary training corpora in the present study.

Fernandez [7] released the Philippine Fake News Corpus, a substantially larger labeled dataset of approximately 21,779 news samples from Philippine digital media. Together with the Fake News Filipino Dataset, this corpus forms the combined primary training resource for model fine-tuning in the present study. Both datasets consist of formal news text, which motivates the need for the supplementary Taglish social media test set to assess real-world generalization.



Cruz and Cheng [2] developed RoBERTa-Tagalog and demonstrated that language-specific pre-training on Filipino and Tagalog corpora yields stronger performance on Filipino NLP tasks than applying multilingual models. Their work also documented the persistent underrepresentation of informal Taglish in available NLP resources — a limitation the present study directly addresses through its novel supplementary dataset.

Montalan et al. [17] introduced Batayan, a holistic Filipino NLP benchmark evaluating large language models across understanding, reasoning, and generation tasks, covering both Tagalog and code-switched Taglish utterances. Their findings revealed substantial performance gaps for Filipino across all evaluated models, highlighting the systematic underrepresentation of Filipino in pre-training corpora and reinforcing the need for specialized evaluation resources. The present study's supplementary Taglish election test set contributes to this growing Filipino NLP evaluation infrastructure.

The reviewed literature confirms that while meaningful progress has been made in Filipino NLP, existing resources remain skewed toward formal text registers. No existing dataset directly supports evaluation of transformer models on informal, code-switched Taglish election content, establishing a clear gap that the present study addresses.

D. Election Misinformation and Political Discourse Online

The study of election misinformation has emerged as a critical research area at the intersection of NLP, political science, and information systems. This section reviews literature on election misinformation, political disinformation on social media, and the computational and empirical approaches developed to detect and analyze it, with particular attention to the Philippine context.

Zubiaga, Liakata, Procter, Bontcheva, and Tolmie [18] examined how users orient to and spread rumors through social media conversational threads, finding that misinformation propagates faster and further than corrections. Their study established that repost structures and threaded conversations play a critical role in misinformation spread — directly relevant to the present study's inclusion of repost captions and eligible threads as units of analysis in the supplementary dataset.

Shu, Mahudeswaran, Wang, Lee, and Liu [19] introduced FakeNewsNet, demonstrating the importance of grounding credibility labels in authoritative fact-checking sources



rather than relying on automated or crowd-sourced annotation. This principle is directly adopted in the present study's annotation procedure, which requires labels to be grounded in verdicts from Vera Files, Tsek.ph, and official COMELEC materials.

Arugay and Baquisal [20] analyzed social media's role in the 2022 Philippine presidential election, documenting how disinformation narratives — including authoritarian nostalgia, conspiracy theories, and democratic disillusionment — spread through social media and contributed to the polarization of the Filipino electorate. Their study provides direct empirical evidence that the 2022 Philippine presidential election was a significant site of social media-driven disinformation centered on specific political figures, directly informing the present study's hypothesis that candidate names and political entities will emerge as influential predictors in Taglish misinformation classifications.

Dewi and Nugroho [21] detected political misinformation in Indonesia's 2024 election using lexicon-based NLP, finding that sentiment-laden and informal language features strongly correlate with misinformation in political social media posts. Conducted in a Southeast Asian context comparable to the Philippines, their study provides empirical support for the present study's focus on informal Taglish expressions as potential influential classification features.

Vera Files [3] and Tsek.ph [4] serve as the primary fact-checking authorities for the Philippine context, providing verified verdicts on election-related claims from the 2022 presidential election period. Both organizations represent the most reliable available resource for grounding the credibility labels in the present study's supplementary dataset.

The reviewed literature establishes that election misinformation is well-documented in the Philippine context, yet existing studies have approached it primarily from political science perspectives with limited engagement with NLP-based detection systems designed for informal, code-switched content. The present study addresses this intersection directly.

E. Model Interpretability in NLP

As transformer models have become the dominant architecture for NLP classification tasks, understanding what drives their decisions has become an essential research concern. Token-level attribution methods have emerged as the primary approach to interpretability in transformer-based NLP. This section reviews foundational and recent

work on attribution methods, with particular attention to Integrated Gradients, SHAP, and their application to text classification.

Sundararajan, Taly, and Yan [22] introduced Integrated Gradients, an axiomatic attribution method that computes each input feature's contribution by integrating gradients along a path from a baseline to the actual input. Integrated Gradients satisfies the completeness axiom — ensuring attribution scores sum to the difference between model outputs on the actual and baseline inputs — providing a theoretically grounded and internally consistent attribution measure. The present study adopts Integrated Gradients as its primary interpretability method for this reason, implemented through the Captum library with a padding token baseline.

Lundberg and Lee [23] proposed SHAP, a unified framework for model interpretation based on Shapley values, satisfying desirable properties including consistency, local accuracy, and missingness. SHAP is adopted in the present study as a secondary cross-validation method to verify the attribution rankings produced by Integrated Gradients, with both methods applied to identify the most reliably influential tokens across all three evaluated transformer models.

Luo, Ivison, Han, and Poon [24] surveyed local interpretation methods for explainable NLP, finding that gradient-based methods such as Integrated Gradients consistently outperform attention-based explanations in faithfulness to model behavior, and that cross-validating multiple attribution methods produces more robust and trustworthy interpretability findings than relying on any single method. This finding directly supports the present study's dual-method interpretability design.

Wu and Ong [25] empirically examined the reliability of BERT explanations across token-level attribution methods, finding that attribution scores vary across model runs and that cross-validation approaches produce more stable explanations. Their findings reinforce the present study's methodological decision to treat only tokens where both Integrated Gradients and SHAP agree as reliable top predictors.

Artstein and Poesio [26] provided a comprehensive treatment of inter-coder agreement metrics, establishing Cohen's Kappa as the standard measure for annotation reliability in computational linguistics. Their work underpins the present study's annotation procedure,



which requires a minimum $\kappa \geq 0.61$ before the supplementary dataset is considered finalized.

The reviewed literature confirms that Integrated Gradients and SHAP are the most theoretically grounded and empirically validated methods for explaining transformer classification decisions, and that cross-validating both methods improves reliability. The present study applies these principles directly, using both methods to identify the top predictors most consistently associated with credible and not credible Taglish election classifications across all three models.

F. Synthesis of the Review of Related Literature

The reviewed literature across all five sections reveals a convergent set of gaps that the present study is specifically positioned to address. Code-switching NLP research has established that Taglish remains underserved by existing models and datasets [2], [12], [14], [15]. Transformer-based fake news detection has demonstrated strong performance in English but lacks comparative benchmarks on code-switched, informal content [8], [16], [9], [5], [10], [11]. Filipino NLP has produced foundational datasets and language models but remains skewed toward formal registers, with no existing resource supporting informal Taglish election content evaluation [6], [7], [2], [17]. The political science literature has documented substantial disinformation activity during the 2022 Philippine presidential election, establishing the real-world relevance of the detection task [18], [19], [20], [21], [3], [4]. Interpretability research has validated Integrated Gradients and SHAP as complementary attribution methods for explaining transformer classification decisions [22], [23], [24], [25], [26].

Together, these bodies of literature establish the technical and contextual foundation for the present study's three-component design: a comparative experimental evaluation, a descriptive corpus analysis, and a model interpretability analysis. The study's contribution lies at the intersection of all five areas, providing both a comparative technical benchmark for Filipino NLP and an interpretable account of the linguistic and political features most strongly associated with Taglish election misinformation.

Chapter III.

METHODOLOGY

A. Research Design

This research adopts a quantitative experimental research design and a supervised machine learning approach on binary text classification of Taglish election misinformation. It uses a comparative model evaluation framework that is implemented based on evaluating three pre-trained transformer models, RoBERTa-Tagalog, mBERT, and XLM-RoBERTa, which are independently fine-tuned on a combined Filipino misinformation corpus. The main goal of the experimental design is to determine under controlled and identical training conditions which model classifies Taglish election-related content as credible or not credible.

Research is both computational and empirical, consisting of dataset preparation, model fine-tuning, performance evaluation, and corpus-based analysis. The evaluation is based on standard classification metrics, including accuracy, precision, recall, and F1-score, and is conducted in two test settings: an internal formal test set which is composed of existing Filipino misinformation datasets, and an additional manually curated Taglish election social media dataset. In addition to the model analysis, a descriptive corpus analysis is performed to examine the linguistic features and thematic patterns of misinformation in the Taglish dataset. This twofold process offers a technical benchmark for transformer models in terms of performance and a detailed context of Taglish election misinformation in the Philippine social media landscape.

B. Dataset Preparation

This study draws data from two distinct sources that serve separate and non-overlapping functions in the research pipeline: (1) a combined primary training corpus sourced from two existing Filipino misinformation datasets, used exclusively for model fine-tuning and internal evaluation; and (2) a supplementary Taglish presidential election test set manually constructed by the researchers from publicly accessible social media content, used exclusively for external model evaluation and top predictor extraction.

1. Primary Training Corpus



The primary training corpus is composed of two existing, publicly accessible Filipino misinformation datasets: the Fake News Filipino Dataset [6] and the Philippine Fake News Corpus [7].

Comprising roughly 3,206 annotated Filipino news items, the Fake News Filipino Dataset [6] utilizes source credibility for its labeling process. Content from authenticated Philippine news agencies is categorized as real or credible, whereas material from sites blacklisted by reputable Philippine fact-checkers is designated as fake or not credible. This collection is available to the public via the authors' official GitHub and HuggingFace repositories.

The Philippine Fake News Corpus [7] contains approximately 21,779 labeled Filipino news samples, similarly labeled by source credibility, and is publicly accessible through its GitHub repository. Together, the two datasets yield an estimated combined corpus of over 24,000 labeled samples, which will serve as the primary supervised learning resource for fine-tuning all three transformer models on binary credibility classification.

The two datasets were selected because they are the most established and widely cited labeled misinformation datasets in the Filipino NLP literature and because their binary labeling structures are compatible with this study's credibility classification task. Both datasets will be accessed in their originally released form, with no modification to individual article content or original labels.

The compatibility of both datasets will be evaluated by examining their structure, text encoding, and label formats before they are standardized into a single tabular format. This unified structure will include, at minimum, the text content, a binary credibility label, and an identifier for the source dataset. To align with the operationalization of this research, label conventions will be harmonized into a single scheme: 1 for credible and 0 for not credible. Following the merger, label distributions will be analyzed to record the class balance of the integrated corpus.

The researchers acknowledge that the primary training corpus is composed mainly of formal Filipino news text. This differs from the informal, code-switched Taglish



discourse found in social media, which is the focus of the supplementary test set. This domain gap is not an oversight but an intentional design feature of the study. By testing models trained on formal Filipino against real-world Taglish election data, the study can evaluate out-of-domain generalization. This approach provides a more rigorous and meaningful assessment of the models' real-world suitability than testing them only on data similar to their training set.

2. Supplementary Taglish Presidential Election Test Set

The supplementary dataset is the novel empirical contribution of this study. It consists of 300 to 500 manually collected, cleaned, and annotated public Taglish social media posts pertaining to the 2022 Philippine presidential election. This dataset will not be used at any stage of model training or validation. It is reserved strictly for external model evaluation and top predictor extraction, maintaining a strict separation between training and evaluation data to prevent leakage.

2.1. Platforms and Collection Window

Posts will be collected from two platforms: Facebook and X (formerly Twitter). These platforms were selected because they are the primary channels through which Filipinos access and share election-related news and commentary, and because Taglish election misinformation during the 2022 Philippine presidential election was extensively documented as circulating on these platforms.

The collection window spans February 8, 2022 to May 9, 2022 — from the official start of the national candidates' campaign period as declared by COMELEC through election day. Post-election content will be excluded. This window was selected as it represents the period of highest election-related social media activity and corresponds to the official campaign period .

2.2. Data Collection Procedure

Posts will be located through platform search functions, keyword-based exploration, public hashtags, and links shared by Philippine fact-checking outlets and election misinformation monitoring reports. Search terms will include presidential candidate names (e.g., Marcos, Robredo, Moreno, Pacquiao, Lacson, de Guzman), election-related terms in Filipino and English (e.g., eleksyon 2022,



boto, pandaraya, dagdag-bawas, COMELEC, halalan, vote fraud, legitimacy), and informal Taglish election expressions and hashtags (e.g., #Eleksyon2022, #Halalan2022, #PHvote2022, sino ang iboboto mo, totoo ba 'to).

No automated scraping tools will be used. All posts will be manually identified, individually reviewed for eligibility, and recorded. This approach was adopted to ensure that each collected post is individually reviewed before entry into the dataset and to maintain compliance with platform terms of service regarding automated data access.

2.3. Eligible Post Types

The primary unit of analysis is a single public social media post or repost caption. A thread on X may be treated as a single textual unit only if it presents a cohesive, self-contained claim that can be evaluated independently without requiring the full thread for comprehension.

For each collected post, the researchers will record the collection date, platform, public URL or post reference, and any visible engagement metrics such as likes, shares, reposts, comments, or reactions where available. These engagement indicators will not be used to determine credibility labels. They will be retained solely for descriptive purposes during the dataset profiling stage and will not factor into model training, evaluation, or predictor extraction at any stage. No private accounts, closed groups, direct messages, or restricted-access materials will be used in the construction of this dataset.

C. Construction and Annotation of the Supplementary Taglish Election Dataset

C.1 Data Preprocessing

Raw data from both the primary training corpus and the supplementary Taglish test set required preprocessing before model fine-tuning and evaluation. Because the two data sources differ substantially in their linguistic register and format, their preprocessing pipelines are described separately below.

1. Primary Corpus Preprocessing

1.1. Merging and Standardization

The data integration process began with the acquisition of the Fake News Filipino Dataset and the Philippine Fake News Corpus from their respective open-access repositories. To ensure structural consistency prior to merging these sources into a consolidated DataFrame, both collections were mapped to a standardized tabular format. This schema maintained essential fields including the raw text content, an indicator for the source dataset, and a binary label for credibility. To align the datasets with the operational definitions of this research, their original labeling systems were harmonized into a single binary classification: 1 representing credible content and 0 representing non-credible content.

1.2. Deduplication

Exact-match deduplication was applied to the text column of the merged corpus to remove identical entries. In addition, the researchers will manually inspect the corpus for repeated articles and near-duplicate entries — such as duplicated headlines differing only in minor punctuation or formatting — and remove them to prevent over-representation of repeated content in training.

1.3. Text Cleaning

The following cleaning steps will be applied to all entries in the primary corpus:

- Removal of HTML tags and markup artifacts introduced during web scraping (e.g., <p>,
,).
- Removal of URLs and hyperlinks.
- Normalization of whitespace and correction of encoding issues, including removal of zero-width characters and standardization of non-standard Unicode characters.
- Removal of excessively short texts (fewer than ten tokens after tokenization), as these are unlikely to contain sufficient linguistic content for meaningful classification [6].
- No removal of Filipino- or Tagalog-specific linguistic markers, particles (e.g., ng, ang, na), or Taglish-specific vocabulary, as these are linguistically meaningful in Filipino NLP.

Cleaning will be deliberately limited in scope to preserve the linguistic content of texts. Stopword removal, stemming, and lemmatization will not be applied to the

primary corpus, as all three transformer models operate on subword tokenization and are designed to process full, unaltered text sequences. Applying these operations prior to tokenization would degrade input representation quality and is inconsistent with standard transformer fine-tuning practice.

2. Supplemental Taglish Test Set Preprocessing

2.1. Deduplication

To prevent the over-representation of repeated misinformation and ensure annotators are not tasked with labeling redundant data, exact duplicates and near-duplicate entries, including minor variations of identical reposted claims, will be detected and eliminated prior to the annotation phase.

2.2. Text Cleaning

The following cleaning steps will be applied to all posts in the supplementary test set following initial screening:

- Removal of platform-specific metadata inadvertently captured during manual collection (e.g., reaction counts, timestamps, and account handles appearing in copied text).
- Removal of URLs, hyperlinks, and embedded media captions not constituting part of the post's textual claim.
- Removal of user mentions (e.g., @username) to prevent models from relying on account identity rather than textual content during classification.
- Removal of non-essential hashtags used purely as metadata labels (e.g., #Halalan2022 appearing as a standalone tag), while retaining hashtags that form part of a readable claim or phrase.
- Emojis and emoticons will be evaluated on a case-by-case basis and removed only when they carry no meaningful contribution to the tone or content of the post.
- Normalization of repeated characters resulting from informal emphasis (e.g., "talaaaaa" normalized to "talaga"), while preserving the root word.
- Correction of spacing inconsistencies, encoding issues, and formatting irregularities common in social media text.



No correction of informal spelling, Taglish code-switching patterns, slang, or non-standard capitalization will be performed. Preserving the authentic informal character of the posts is essential to the study's goal of evaluating real-world model performance on Taglish text, and altering these features would compromise the ecological validity of the supplementary test set and undermine the interpretability analysis. Any post that becomes inaccessible or unverifiable during this stage will be excluded from the final dataset.

C.2 Annotation Procedure

The annotation procedure covers the labeling of the supplementary Taglish presidential election test set and the supplementary metadata tagging applied to each post. This stage is conducted after data collection and preprocessing and before model evaluation, ensuring that all credibility labels are finalized prior to any contact with the models.

1. Annotation Guide and Labeling Rules

Each retained post will be independently labeled by two members of the research team as either **credible** or **not credible**, using a predefined annotation guide grounded in verdicts and supporting evidence from Vera Files, Tsek.ph, and official COMELEC statements. Annotators are required to assess the factual accuracy of the post's claim, not the political alignment, tone, or candidate identity of any figure mentioned. The following labeling rules will apply:

- **Credible (Label = 1).** The post contains a factual claim that is supported or confirmed by reliable and verifiable sources.
- **Not Credible (Label = 0).** The post contains a factual claim that is false, misleading, manipulated, unsupported, or contradicted by reliable and verifiable sources.
- **Verifiable Claim Requirement.** Only posts containing a factual claim checkable against documentary or authoritative evidence will be labeled.
- **Evidence Basis.** Annotators must ground their labels in credible references such as COMELEC statements, verified news reports, Vera Files, and Tsek.ph.



- **Main-Claim Rule.** Labeling will be based on the primary election-related factual claim expressed in the post, regardless of whether secondary statements are also present.
- **Partially True or Misleading Content.** Posts that contain factual elements but frame them in a way that distorts public understanding will be labeled as not credible within the binary framework [19].
- **Satire, Parody, and Opinion.** Posts that are clearly satirical, parodic, or purely opinion-based will not be labeled and will be excluded [27].
- **Ambiguous Cases.** Claims that are unclear, incomplete, or insufficiently contextualized for conclusive verification will be marked for further review.
- **Disagreement Resolution.** In cases of annotator disagreement, a third member of the research team will serve as tiebreaker, reviewing the post and the relevant fact-check evidence independently before issuing a final label.
- **Unresolved Cases.** Posts that remain unresolved after third-annotator review will be excluded from the final dataset.

The researchers acknowledge that using members of the research team as primary annotators is a practical constraint of the study and introduces the possibility of familiarity bias. This limitation is documented transparently, and the inter-annotator agreement procedure described below is intended to mitigate its effect by enforcing independent labeling before any discussion or resolution occurs.

2. Inter-Annotator Agreement

Inter-annotator agreement will be measured using **Cohen's Kappa (κ)** [26]. A minimum agreement threshold of $\kappa \geq 0.61$, corresponding to substantial agreement, must be achieved before the supplementary dataset labels are considered finalized. If agreement falls below this threshold, the annotation guide will be reviewed and revised before relabeling proceeds. The final Cohen's Kappa score will be reported transparently in the results chapter alongside the finalized dataset's composition statistics.



C.3 Entity and Issue Tagging

In addition to assigning a credibility label, annotators will record supplementary metadata tags for each post capturing the major political entities, candidate names, election institutions, and issue categories present. This tagging layer supports the corpus profiling stage and the interpretability analysis by allowing model-generated top predictor rankings to be compared against human-coded dataset themes, producing a more interpretable and externally validated account of what the models are responding to.

The following fields will be recorded for each post where applicable:

Field	Examples
Primary candidate / political actor	Marcos, Robredo, Pacquiao, Moreno, Lacson
Institution mentioned	COMELEC, courts, election-related bodies
Issue category	vote fraud, false endorsement, historical claim, election legitimacy, misleading statistics
Claim type	false quote, fabricated event, manipulated context, unsupported allegation
Language pattern	Taglish slang, hashtag, meme phrase, balanced code-switching

The final output of this stage will be a clean, manually verified supplementary Taglish presidential election test set of 300 to 500 samples, reserved exclusively for model evaluation and predictor extraction and never used for model training.

D. Dataset Profiling and Exploratory Corpus Analysis

Dataset profiling will be conducted after the supplementary Taglish presidential election test set has been collected, screened, cleaned, and annotated. This stage aims to describe the overall composition of the dataset before it is used for model evaluation and interpretability analysis. Since the supplementary dataset serves as the study's real-world



evaluation set, profiling is necessary to establish a clear overview of its credibility labels, platform sources, candidate mentions, issue categories, claim types, and Taglish language patterns.

The researchers will summarize the dataset using frequency counts, percentages, and descriptive tables. The profiling process will include the distribution of credible and not credible posts; the number of posts collected from Facebook and X; the frequency of presidential candidates and political entities mentioned; the common election-related issue categories; the types of claims found in a post; and the observed code-switching or informal Taglish expressions. Public engagement indicators such as likes, shares, comments, reposts, or reactions may also be recorded when visible, but these will only be used for descriptive context and not as model input or labeling criteria.

The results of dataset profiling will serve as the empirical baseline for the succeeding analyses. It will help the researchers understand the structure of the supplementary dataset and provide context for interpreting the model evaluation results and top predictor rankings.

Profiling Category	Description
Credibility label distribution	Summarizes the number and percentage of credible and not credible posts.
Platform distribution	Summarizes how many posts came from Facebook and X.
Candidate and political entity distribution	Summarizes how frequently candidates, institutions and political actors appear in the dataset
Issue and claim distribution	Summarizes the common issue categories and claim types found across the dataset.
Taglish language pattern distribution	Summarizes observed code switching patterns, slang, hashtags and informal expressions.



Engagement indicator summary	Summarizes visible likes, shares, comments, reposts, or reactions for descriptive context only.
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E. Model Fine-Tuning

This section describes the model training and evaluation pipeline used in this study. Three pre-trained transformer models are independently fine-tuned for binary credibility classification (credible / not credible) and evaluated under identical conditions to ensure a controlled and fair comparison.

E.1 Transfer Learning Paradigm

This study adopts a transfer learning approach in which pre-trained transformer models are fine-tuned on a domain-specific classification task rather than trained from scratch. This is well-suited to low-resource NLP settings, such as Taglish misinformation detection, where labeled domain-specific data is limited. Each pre-trained model has already acquired general linguistic representations through large-scale pre-training; fine-tuning adapts these representations to the binary classification task using a comparatively small, labeled corpus.

A classification head — consisting of a dropout layer followed by a linear projection layer — is attached to each model's pooled output representation. This head maps the model's final hidden state to two output logits corresponding to the binary target classes: credible (label = 1) and not credible (label = 0). Both the classification head and the transformer's pre-trained weights are updated during fine-tuning via backpropagation using cross-entropy loss.

E.2 Model Selection Rationale

Three transformer models were selected to represent different pre-training strategies relevant to Taglish:

1. RoBERTa-Tagalog

- Pre-trained primarily on Filipino and Tagalog corpora.



- Selected as the language-specific baseline — its training data most closely matches the target language of this study [2].
- Represents the language-specific pre-training condition in this comparison.

2. mBERT (bert-base-multilingual-cased)

- Pre-trained on Wikipedia text from 104 languages, including Filipino.
- Selected to represent a broad but shallower multilingual baseline with limited Tagalog-specific depth [8].
- May leverage shared multilingual representations to handle Taglish's English components.

3. XLM-RoBERTa (xlm-roberta-base)

- Pre-trained on a larger and more balanced multilingual Common Crawl corpus across 100 languages [16].
- Consistently outperforms mBERT on cross-lingual NLP benchmarks.
- Serves as the stronger multilingual baseline in this comparison.

Together, these three models enable a direct comparison of language-specific pre-training depth versus multilingual pre-training breadth for Taglish misinformation detection.

E.3 Training Configuration and Fairness Conditions

To ensure that performance differences reflect genuine model capability rather than experimental conditions, all three models are fine-tuned under the same dataset split and hyperparameter configuration.

Dataset Split:

Subset	Source	Proportion
Training	Combined Filipino fake news corpus [6],[7]	70%
Validation	Same corpus	15%
Internal Test	Same corpus	15%



External Test	Supplementary Taglish social media test set (300–500 manually labeled posts)	Held out entirely
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The supplementary Taglish test set is withheld from all training and validation stages to prevent data leakage and to ensure that performance on this set reflects genuine generalization to real-world Taglish content.

Hyperparameter Configuration:

Hyperparameter	Value / Setting
Optimizer	AdamW with linear warmup scheduling
Learning rate	2e-5
Batch size	16 or 32 (held constant across all models; set by GPU memory)
Training epochs	3–5, with early stopping on validation loss plateau
Max sequence length	128 tokens
Loss function	Cross-entropy loss
Dropout rate	0.1 (classification head)
Hardware	Google Colab Pro with GPU acceleration

The checkpoint with the lowest validation loss across all training epochs is selected for evaluation. Minor architectural adjustments required by a specific model (e.g., tokenizer vocabulary differences) will be documented in the training log. Where full uniformity cannot be maintained without introducing technical incompatibilities, batch size is adjusted based on GPU memory constraints while keeping all other hyperparameters constant.

E.4 Model Fine-Tuning Procedure

Fine-tuning proceeds in the following sequence for each of the three models:

- The public library Hugging Face Transformers, which includes RoBERTa-Tagalog, bert-base-multilingual-cased (mBERT), and xlm-roberta-base (XLM-RoBERTa) is utilized to load all pre-trained models and their tokenizers.
- In each model, tokenize the training corpus texts to output of the input IDs along with attention masks using its native tokenizer. Each sequence is padded or truncated to a maximum of 128 tokens. Tokenization differences—RoBERTa-Tagalog uses BPE, mBERT uses WordPiece, XLM-RoBERTa uses SentencePiece—are intrinsic to each model's design and are not treated as fairness violations.
- A classification head (dropout + linear layer) is attached to each model's pooled output and initialized with random weights prior to fine-tuning.
- For the specified number of epochs, each model is trained on the training subset. The training loss, validation loss and validation accuracy of the three models are recorded at each epoch.
- The minimum validation loss checkpoint is saved and used for all subsequent evaluation and interpretability analysis.

F. Model Evaluation

F.1 Two-Level Evaluation Structure

Each model is evaluated on two distinct test environments to assess both in-distribution performance and real-world generalization:

Level 1 — Internal Formal Test Set:

- The 15% held-out split from the combined Filipino news corpus.
- Consists of formal Filipino news articles from the same source distribution as training data.
- Provides a controlled performance baseline measuring how well each model learned to classify formal Filipino misinformation text.

Level 2 — Supplementary Taglish Social Media Test Set:

- 300–500 manually annotated, publicly accessible social media posts in Taglish, collected from Facebook and X.

- Represents a fundamentally different distribution from the training data: informal, code-switched, real-world content.
- Constitutes the primary measure of each model's real-world generalization capability.

This two-level evaluation structure is necessary because strong performance on formal in-distribution data does not guarantee strong performance on informal social media content. The performance gap between the two test levels is treated as key evidence of each model's robustness to domain shift from formal Filipino news to informal Taglish content.

F.2 Evaluation Metrics

Performance is gauged using typical NLP classification measures computed from each model's predictions on two test sets. The not credible class (label = 0) is treated as the positive class, as misinformation detection is the primary task of interest.

- Accuracy - the overall percentage of correctly classified samples. Treated as a secondary indicator due to possible class imbalance.
- Precision - of all posts flagged as not credible, how many were truly misinformation.
- Recall - of all actually not credible posts, how many were correctly identified. Higher recall means fewer missed misinformation cases.
- F1 - harmonic mean of precision and recall. Both macro-averaged and weighted F1-scores are reported; weighted F1-score serves as the primary basis of model comparison, given potential class imbalance in the dataset.
- Confusion matrix - displays counts of true positives, true negatives, false positives, and false negatives per model per test set for full transparency.

F.3 Comparative Results Presentation

Each model's performance metrics — accuracy, precision, recall, macro F1-score, and weighted F1-score — are presented side-by-side in a comparative results table for both the internal formal test set and the supplementary Taglish test set. The



best-performing model on the supplementary test set by weighted F1-score is identified as the primary model for deeper interpretability analysis.

Where the best-performing model on the internal formal test set differs from the best-performing model on the supplementary Taglish test set, this discrepancy is treated as a primary finding — direct evidence that formal Filipino news performance does not reliably predict real-world Taglish generalization. This comparison forms the core empirical contribution of the evaluation component.

F.4 Statistical Significance Testing

Performance differences between models will not be interpreted on the basis of metric values alone. To determine whether observed differences in classification performance are statistically meaningful rather than attributable to chance, McNemar's test will be applied to pairwise comparisons of model predictions on both test sets [28]. A significance level of $\alpha = 0.05$ will be used. Only differences that pass this threshold will be characterized as meaningful performance advantages in the comparative discussion. This step is particularly important on the supplementary Taglish test set, where the sample size is smaller and metric differences may be more susceptible to variance.

G. Top Predictor Extraction and Model Interpretability Analysis

G.1 Interpretability Method

Following evaluation, a token-level interpretability analysis is conducted on the supplementary Taglish test set to identify which textual features most strongly influence each model's classification decisions. Two complementary attribution methods are used:

Primary Method — Integrated Gradients (via Captum library):

- Computes the attribution score of each input token by integrating the model's gradients from a baseline input (padding tokens) to the actual input.
- Satisfies the completeness axiom — the sum of all token attribution scores equals the difference between the model's output on the actual input and its



output on the baseline — providing a theoretically grounded attribution measure.

- Applied to all posts classified as not credible from the supplementary test set.

Secondary Method — SHAP (SHapley Additive exPlanations):

- Applied to a stratified subsample of the interpretability subset (minimum 30% of not credible predictions, with an absolute floor of 50 posts) to cross-validate Integrated Gradients rankings.
- The subsample is stratified across candidate mentions, issue categories (health misinformation, electoral procedure, character attacks, policy misrepresentation), and collection periods to ensure representativeness.
- Where both methods consistently rank the same tokens highly, those tokens are treated as robust top predictors. Disagreements between methods are reported as a limitation.

Subword tokens are merged and their scores aggregated before ranking to ensure results reflect meaningful linguistic units rather than tokenization artifacts. A minimum frequency threshold of five posts is applied to prevent low-frequency tokens from dominating rankings. Attribution results are organized into categories: political names and entities, election-related terms, Taglish expressions, issue-related keywords, and meme or viral phrase references.

G.2 Predictor Extraction Procedure

The interpretability analysis will be applied to each model's predictions on the supplementary Taglish presidential election test set. The primary focus will be on posts classified as not credible, as these are most directly relevant to the study's misinformation detection objective. Posts classified as credible will be analyzed separately as a contrast set to identify which features distinguish misinformation from credible election content, providing a more complete picture of how each model draws the boundary between the two classes.

Because transformer tokenizers decompose text into subword units, subword tokens will be merged and interpreted at the word or phrase level where appropriate to improve the readability and interpretability of predictor rankings.

The extraction procedure will proceed as follows. First, for each post in the interpretability subset, token-level attribution scores will be computed using Integrated Gradients with the padding token baseline defined in Section G.1. Primary analysis will focus on posts classified as not credible. Second, attribution scores will be aggregated across all not credible posts by averaging scores per unique token across the posts in which each token appears. Credible posts will be aggregated separately for contrast analysis. Third, tokens will be ranked by their aggregated attribution scores to produce a list of top predictors per model. Fourth, a minimum frequency threshold will be applied to filter out tokens appearing in fewer than 5% of posts in the not credible subset, ensuring that top predictors reflect consistent patterns across the dataset rather than isolated or post-specific artifacts. This threshold will be reviewed and adjusted if the final dataset size or token distribution warrants revision, and any adjustment will be documented and justified. Fifth, the ranked predictor list will be organized into the following interpretable categories: political names and entities, election-related terms, Taglish expressions and slang, hashtags and meme phrases, and issue-related keywords.

The output of this procedure will be a ranked list of top textual predictors per model, each associated with an aggregated attribution score reflecting its overall influence on that model's misinformation classifications across the supplementary test set.

G.3 Cross-Model Predictor Comparison

The ranked predictor lists from all three models will be compared to examine the degree of agreement and disagreement in what each model identifies as influential features of Taglish election misinformation. This comparison will address the following dimensions.

Overlapping predictors. Tokens or terms appearing among the top predictors of two or more models suggest consistent linguistic or political signals that are robust across different pre-training strategies and model architectures.



Model-specific predictors. Tokens appearing as top predictors in only one model reveal differences in how language-specific and multilingual pre-training strategies process Taglish code-switched content — a finding relevant to decisions about model selection for Filipino NLP applications.

Political name patterns. Whether specific candidate names such as Marcos, Robredo, Pacquiao, or Duterte consistently appear as top predictors across models, and whether their presence is concentrated within particular issue categories or claim types identified during annotation in Section D.3.

Taglish-specific patterns. Whether Taglish expressions, slang, hashtags, or meme phrases appear among the top predictors, and whether these features are captured more effectively by the language-specific RoBERTa-Tagalog or by the multilingual models — an empirical question that bears directly on the study's model selection rationale in Section E.2.

G.4 Validation Against Corpus Findings

The model-generated top predictor rankings will be compared against the corpus-level findings from Section C — specifically the frequent terms and n-grams, political entity frequencies, and issue category distributions identified during dataset profiling. This comparison functions as a cross-validation between human-coded corpus patterns and machine-generated attribution scores.

Where model-identified top predictors align with corpus-level frequent terms and annotator-coded entity tags from Section D.3, this convergence strengthens the interpretation that those features represent meaningful signals within the collected Taglish election dataset rather than artifacts of a particular model's behavior. Where significant divergence exists between corpus patterns and model predictors, this will be treated as a substantive finding in its own right rather than dismissed as noise. Such divergence may indicate that certain models are responding to distributional patterns in the data that are not immediately visible through manual frequency analysis — a possibility that is analytically interesting and will be discussed accordingly.

H. Model Demonstration (Simulation)

This section describes the simulation-based demonstration environment used to operationalize the three fine-tuned transformer models in a controlled setting. Instead of deploying a production system, the study implements a local inference-based simulation that allows comparative observation of model behavior on identical inputs. This component serves as an extension of the evaluation framework by enabling instance-level analysis of classification outputs.

H.1 Purpose and Scope of the Simulation

The simulation is designed to support the interpretability of the quantitative evaluation results by enabling direct comparison of model predictions on the same input text. While Section F evaluates model performance using aggregate metrics, the simulation allows observation of individual prediction behavior, including differences in classification decisions and confidence outputs across models.

The primary purpose of this component is comparative demonstration. It enables analysis of how RoBERTa-Tagalog, mBERT, and XLM-RoBERTa respond to the same Taglish input at the instance level, complementing the dataset-level evaluation performed in the previous sections.

This simulation is strictly limited to research purposes and does not function as a deployed or publicly accessible application.

H.2 System Design and Inference Process

The simulation loads the trained checkpoints of RoBERTa-Tagalog, mBERT, and XLM-RoBERTa from local storage. All models are used in inference mode, with training disabled during execution.

A single Taglish text input is processed per inference cycle. The input is tokenized using the corresponding tokenizer of each model and converted into input IDs and attention masks. The tokenized sequence is then passed through each model to generate output logits.

Each model produces:

- A binary classification label (credible or not credible) based on the highest logit score
- A confidence score derived from the softmax probability distribution over the output classes

The models operate independently but are evaluated on identical inputs to ensure controlled comparison.

H.3 Attribution Computation

To complement classification outputs, token-level interpretability is computed using Integrated Gradients via the Captum library. This method is applied to each input to estimate the contribution of individual tokens to the final prediction.

Attribution values are computed relative to a baseline input consisting of padding tokens, consistent with the approach used in the interpretability analysis in Section G.1. The resulting token attributions are aggregated at the subword level to improve interpretability and reduce tokenization artifacts.

SHAP is not applied in the simulation due to its higher computational cost, which makes it unsuitable for per-input computation in a real-time inference context.

H.4 Operational Constraints and Limitations

The simulation is designed strictly as a controlled evaluation tool and is subject to several limitations.

First, it does not reflect production-level system constraints such as scalability, concurrency handling, or deployment latency. The results are dependent on local hardware conditions and are not representative of deployed system performance.

Second, all outputs—including predicted labels and attribution scores—are derived from models trained on a specific Filipino misinformation dataset. As such, results should be interpreted as model behavior under learned representations rather than definitive classifications of truth or falsehood.

Third, instance-level attribution results generated during simulation are not directly comparable to aggregate attribution statistics presented in Section G.1, as they reflect single-input explanations rather than corpus-level patterns.

Despite these limitations, the simulation provides a controlled environment for observing comparative model behavior at the input level, supporting the broader evaluation objectives of the study.

I. Fairness, Validity and Ethical Considerations

This section delineates the fairness, validity, and ethical considerations integrated into the research pipeline. Due to the politically sensitive nature of the 2022 Philippine presidential election and the implications of findings that link specific candidates to misinformation, the study incorporates rigorous safeguards across annotation, data processing, and research conduct. These measures are designed to ensure that all empirical outcomes remain transparent, accountable, and methodologically robust.

1. Annotation Validity and Political Fairness

For the credibility assessments to remain objective and thorough, all verifications of posts in the supplementary Taglish presidential election test set would be based solely on verifiable factual claims. All facts would be corroborated using official statements made by the COMELEC and the results of fact checks made available at Tsek.ph and Vera Files. Annotators would be explicitly instructed to focus on the factual accuracy of the primary claim made in the post in question without consideration of any other aspects, including the politician/candidate the post refers to, their political party affiliation, rhetorical nature, or general tone. As such, a post could be considered not credible in case of an existing specific false claim either contradicted by government documents or found to be fake by an accredited Philippine fact-checking organization.

Drawing from the work of Potthast et al. [27], all posts containing rhetoric and political opinions or satire not accompanied by an objective, verifiable factual claim would be disregarded at the screening stage. This exclusion rule applies equally to posts about all political figures and candidates present in the data set in order to avoid possible systematic over-labeling of content produced by certain actors.

Inter-annotator agreement will be calculated using Cohen's Kappa (κ) measure following the established guidelines of Artstein and Poesio [26]. A value of $\kappa \geq 0.61$ will be needed in order to include the label in the final dataset. If the consensus is not reached even after mediation by a third annotator, the post will be removed from the



study to ensure no single person will have control over labeling. Additionally, the credibility rubric used by this research for analysis will be published as a supplemental outcome in order to encourage external assessment of the research method.

It should be recognized that using the team members as primary annotators limits the scope of choice. Nevertheless, an agreement protocol involving independent labeling and subsequent discussion is employed in order to neutralize the potential effect.

2. Data Privacy and Platform Ethics

The supplementary Taglish presidential election test set will be constructed exclusively from publicly accessible social media content. No private accounts, closed groups, direct messages, or restricted-access materials will be used at any stage of data collection. All posts will be manually identified and individually reviewed for eligibility prior to entry into the dataset, ensuring compliance with platform terms of service regarding automated data access. No automated scraping tools will be used.

Engagement metrics such as likes, shares, comments, reposts, and reactions will be recorded solely for descriptive purposes during the dataset profiling stage and will not be used as model input, labeling criteria, or inferential evidence at any stage of the study. User mentions appearing in collected post text will be removed during preprocessing to prevent models from relying on account identity rather than textual content during classification and to minimize the risk of associating individual social media users with misinformation findings.

The study does not seek to identify, expose, or make claims about the individuals who authored or shared the collected posts. The unit of analysis is the post content itself, not the behavior or identity of the posting user.

3. Research Integrity and Scope of Claims

In addition to this limitation, this study does not make any conclusions about the intent behind, the origin of, or the political benefactors of any of the misinformation that will be identified in the supplementary dataset. The findings of this study will



solely focus on the distribution of credibility levels in relation to the categories within which candidate mentions appear based on the results of annotation and model classification within the conditions set forth in this study. Researchers will distinguish between what can be concluded from this study's findings in terms of observed patterns and what cannot be claimed, namely, causality, intent, and electoral manipulation.

This study is purely academic in nature and is being performed without commercial interests in mind. The methodology, cutoffs, criteria, and processes used in this study are all set out before data collection and analysis takes place in order to ensure that there is no manipulation of the research process in reaction to what was found in the data. The minimum candidate level sample size, the Cohen's kappa agreement cutoff, and the train-validation-test split ratio will all be pre-registered in this methodology chapter.

Researchers recognize that politically charged findings may potentially come under outside scrutiny but are confident that their rigorous methodology and the fact that all labeling occurs on the basis of factual information and pre-publication of the annotation guidelines can stand up to scrutiny.

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